ITAI-1371 - Intro to Machine Learning

**Individual Contribution Journal**

**Camila Ferreira da Silva**

My contribution to Lab 9 included reviewing the ensemble-method concepts in Part 1 and checking the setup in Part 2. I organized the imports and Iris dataset code and verified the 70/30 training-testing split. Because the code cell did not display output automatically, I added print statements to check the dataset sizes. The output showed 150 observations total, with 105 in the training set and 45 in the testing set. I also integrated the group's completed sections and ran the notebook from top to bottom to check that the cells executed and produced the expected outputs.

**Danh Nguyen**

I was responsible for the Random Forest model (Part 4). I trained a Random Forest with 100 trees on the Iris training data, evaluated its accuracy on the test set, and compared it with the single Decision Tree. Because both models tied at 100% on the test set, I added a 10-fold cross-validation comparison (Random Forest 96.0% vs. Decision Tree 95.3%) and wrote the explanation of the results.

**Great Inioluwa Olayiwola**

For this lab, I built and analyzed both Decision Tree and Random Forest classifiers in order to classify Iris flower types using Scikit-Learn. I also wrote Python code for instantiating and training both models using a 70/30 split and evaluated the models with accuracy_score. I then created and plotted a horizontal bar graph to illustrate the feature_importances_ extracted from the trained Random Forest. From this, it was clearly shown that the petal length and petal width were the main features utilized in the decision process of the classification.

**Karima Sadruddin**

I worked on Part 3 of the Ensemble Methods lab, where I trained and evaluated a single Decision Tree model using the Iris dataset. I used DecisionTreeClassifier to train the model, made predictions on the test data, and calculated its accuracy using accuracy_score. The Decision Tree achieved 100% accuracy, correctly classifying all 45 test samples. Through this activity, I learned how to train a machine learning model, evaluate its performance, and understand how a single Decision Tree makes predictions.

**Quan Thach**

For this lab, I worked with my group on the full Module 09 notebook. I helped set up the Iris data and the train and test split, and I went through the single Decision Tree and Random Forest sections so we understood what each model was doing before we compared them. I also helped read the accuracy results and the feature importance plot, and I wrote answers for the Knowledge Check questions based on what we saw. Along the way, I checked the explanations with my teammates so that our final answers matched the results in the notebook.